ITNPA – 14
Formative Assessment

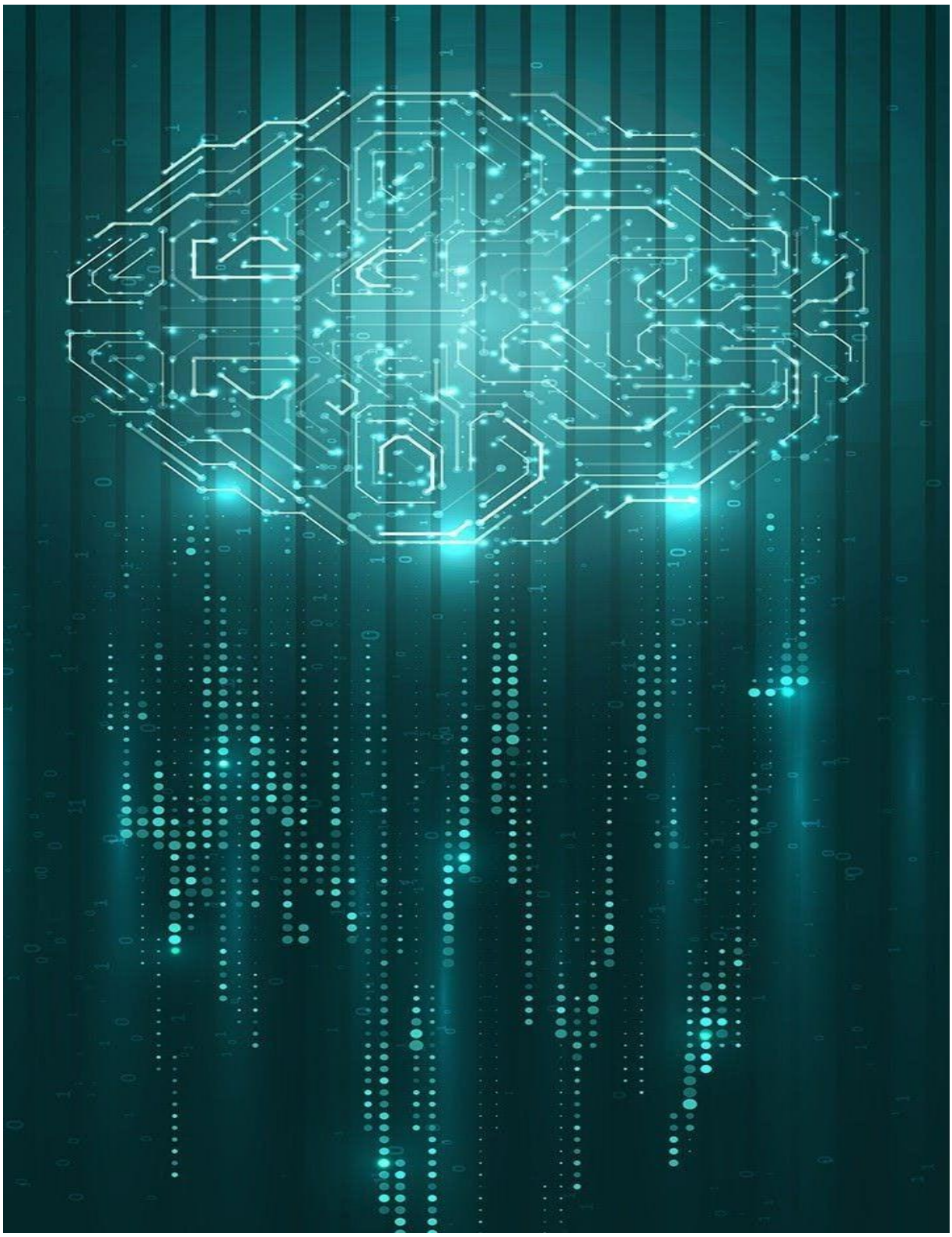


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Question 1

1.1) Please see attached Jupyter script.

The data contains 1,138 abstracts (900 classical and 238 AI), so the classes are imbalanced (as seen in Figure 1). Noise such as copyright notices, URLs, numbers, and punctuation were found (Figure 2) and removed. In addition to this, 13 duplicates were dropped, the text was tokenised, and stopwords were removed and the words were lemmatised with NLTK.

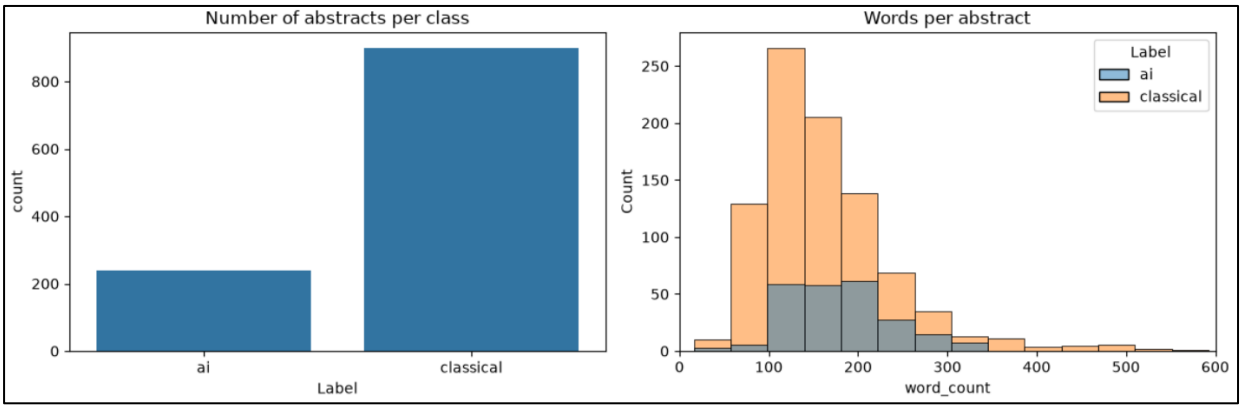


Figure 1: Number of abstracts per class and distribution of words per abstract.

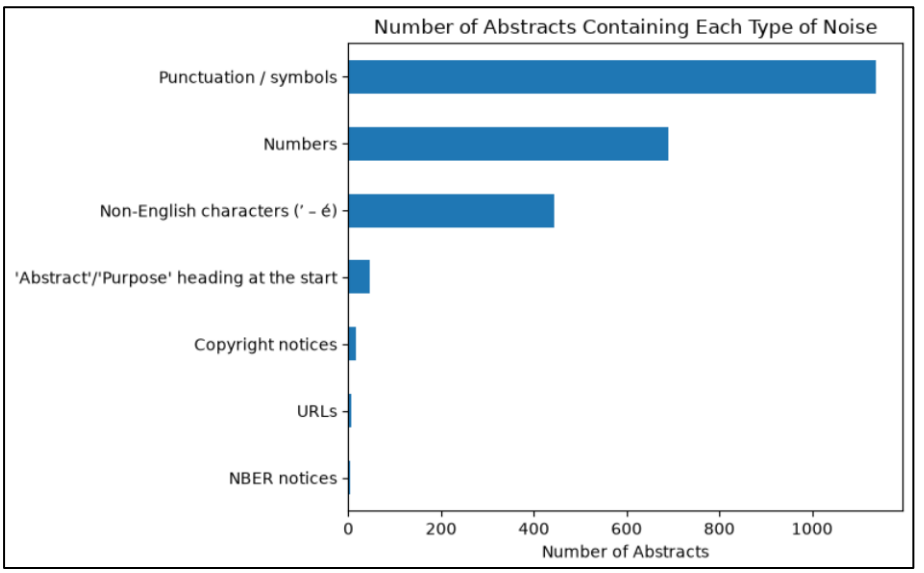


Figure 2: Number of abstracts containing each type of noise.

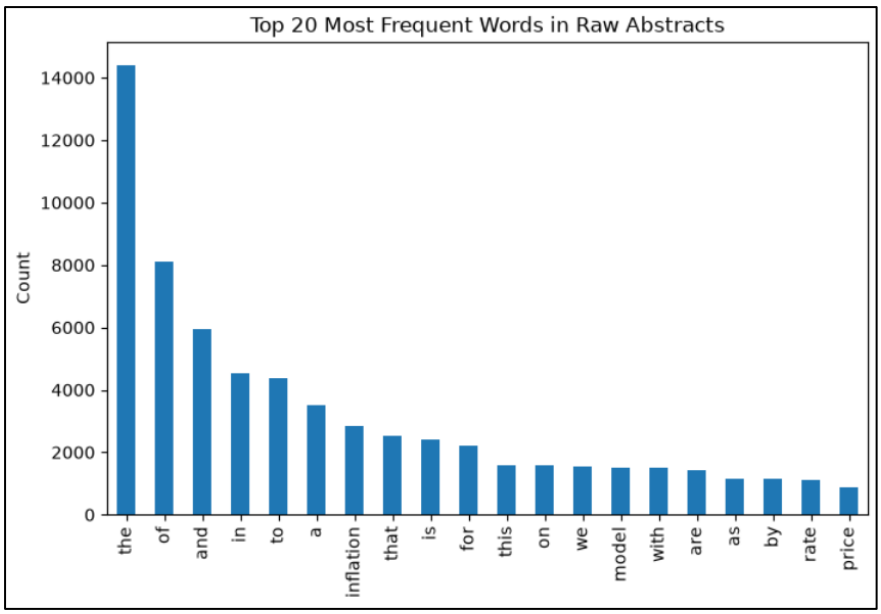


Figure 3: Top 20 most frequent words in the raw abstracts.

1.2) Please see attached Jupyter script.

The cleaned data was saved to cleaned_abstracts.csv and the label (AI or Classical) was encoded with LabelEncoder. TF-IDF features were created, and a chi-square test showed that AI terms such as 'machine learning' are the most useful features (Figure 4).

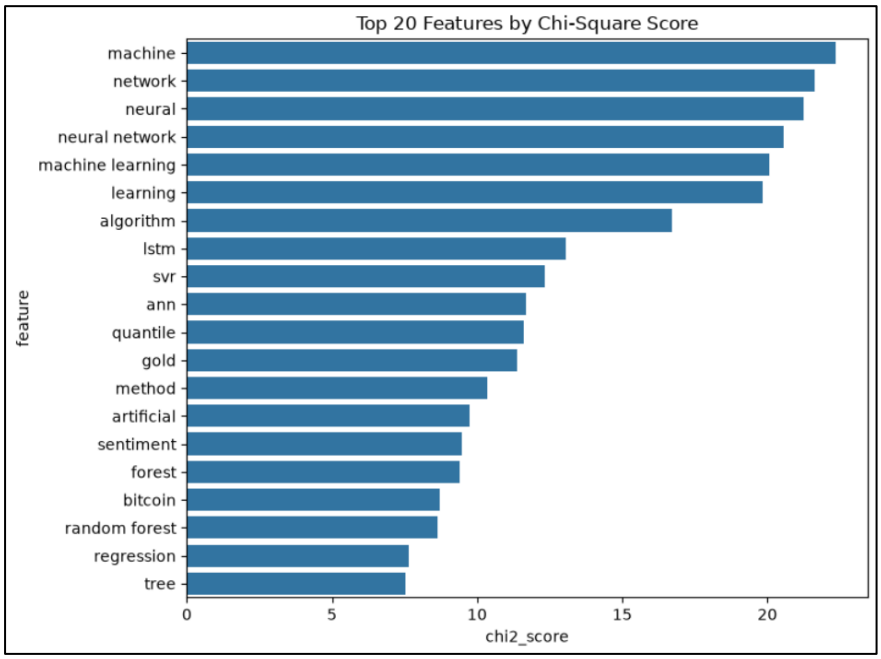


Figure 4: Top 20 TF-IDF features by chi-square score.

1.3) Please see attached Jupyter script.

The logistic regression baseline reached 83.6% test accuracy but only found 21% of AI abstracts, with a macro-F1 of 0.63 (Figure 5) and a 5-fold CV macro-F1 of 0.640 (scikit-learn developers, 2026). The learning curve (Figure 6) shows that the model underfits.

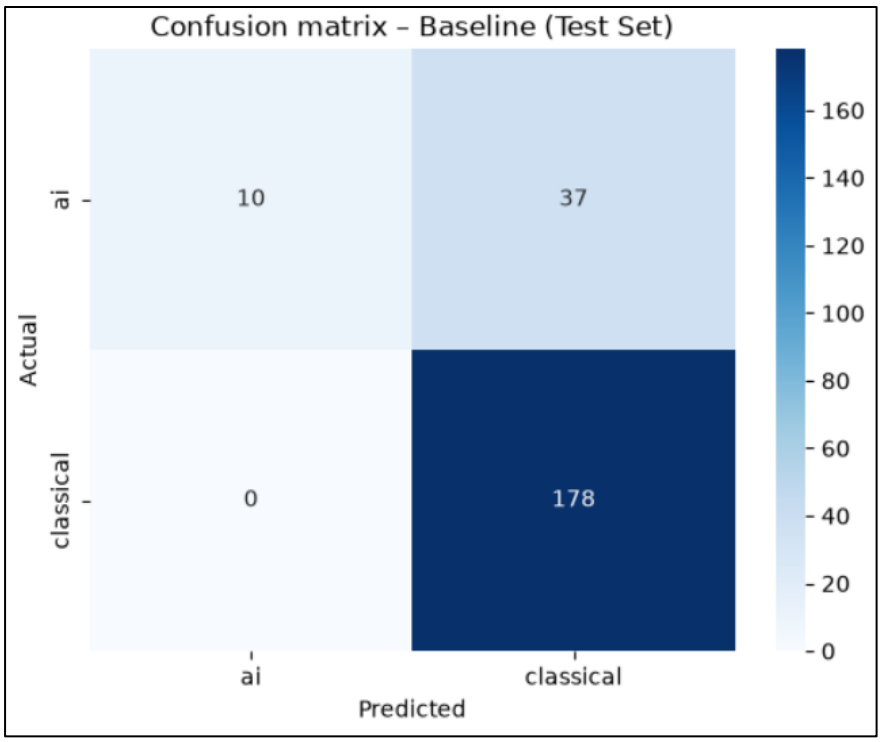


Figure 5: Confusion matrix of the baseline Logistic Regression model (test set).

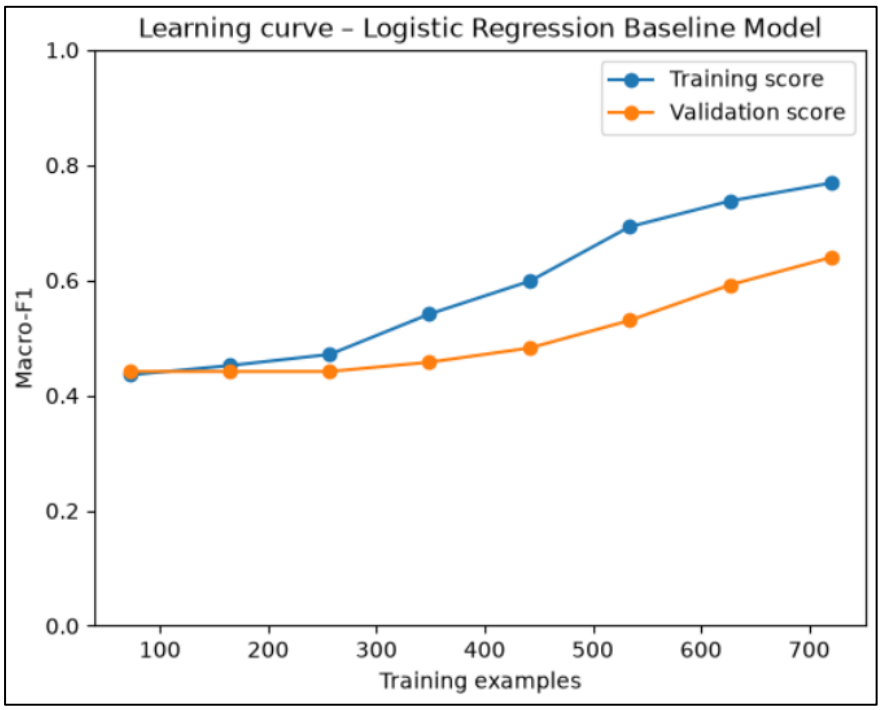


Figure 6: Learning curve of the baseline Logistic Regression model (macro-F1).

Question 2

2.1) Please see attached Jupyter script.

A linear SVM, Random Forest, Gradient Boosting, and Complement Naïve Bayes were compared with the baseline. All of them outperformed the baseline, and the Linear SVM had the best test macro-F1 of 0.860 (scikit-learn developers, 2026) (Figure 7).

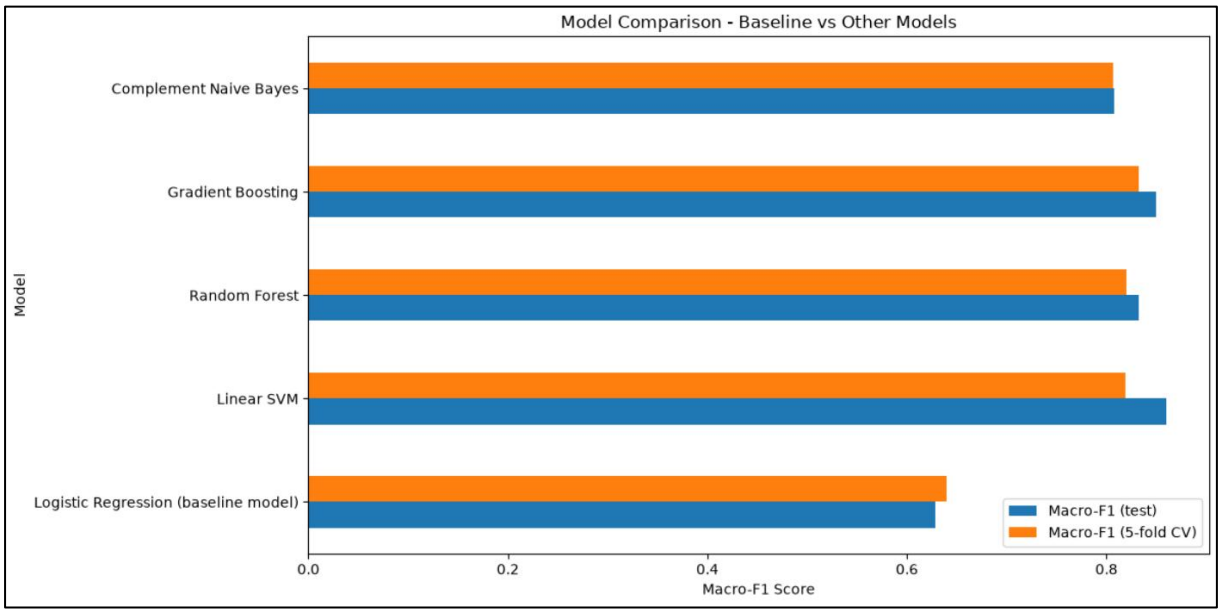


Figure 7: Macro-F1 comparison of the baseline and other models (test set and 5-fold CV).

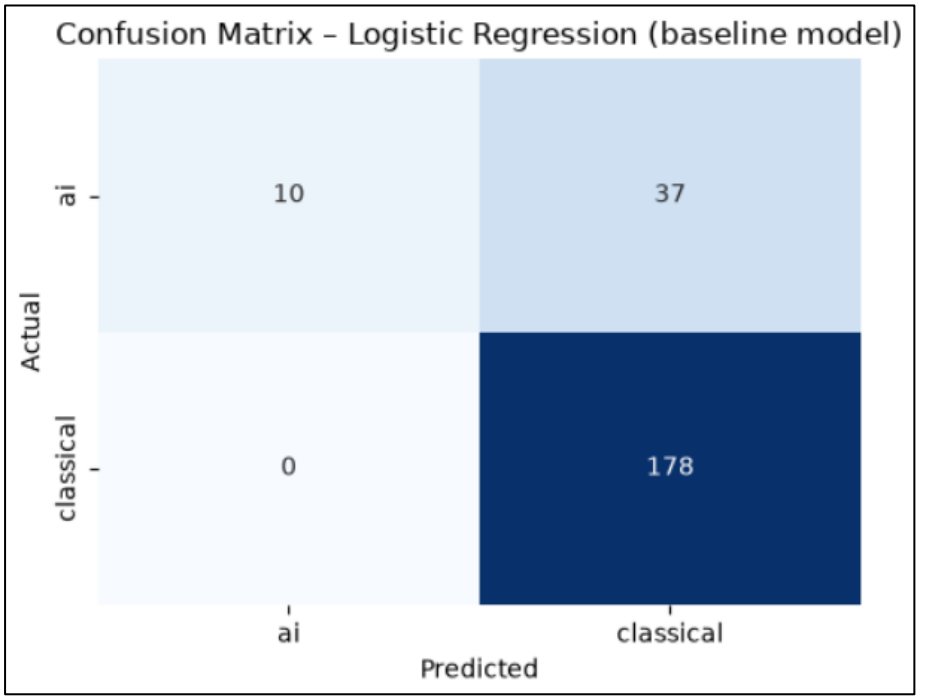


Figure 8: Confusion matrix – Logistic Regression (baseline model)

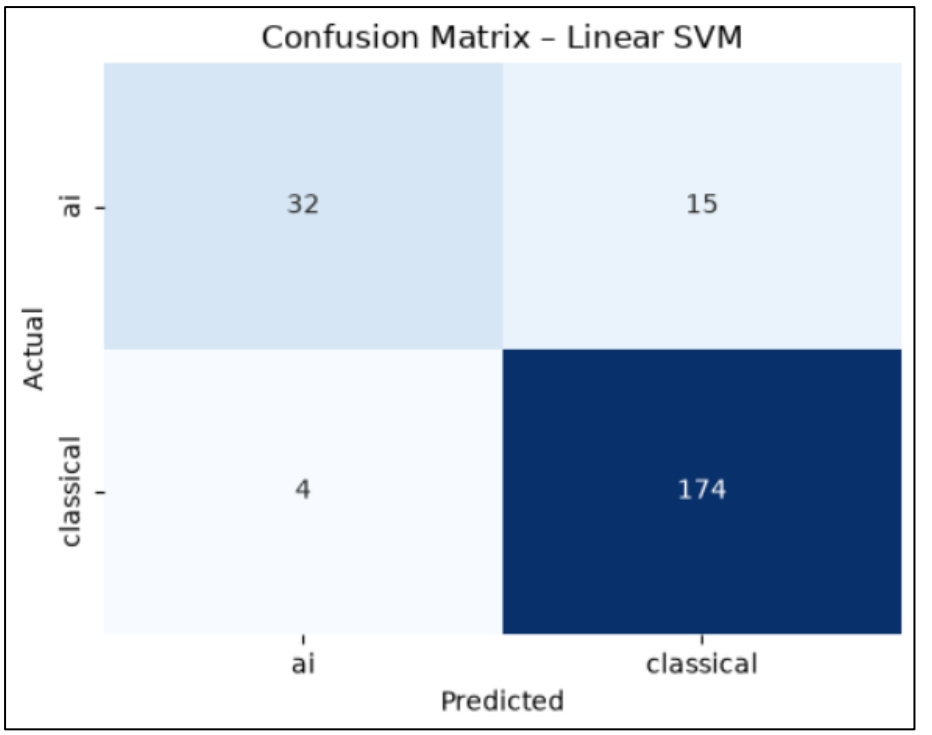


Figure 9: Confusion matrix – Linear SVM

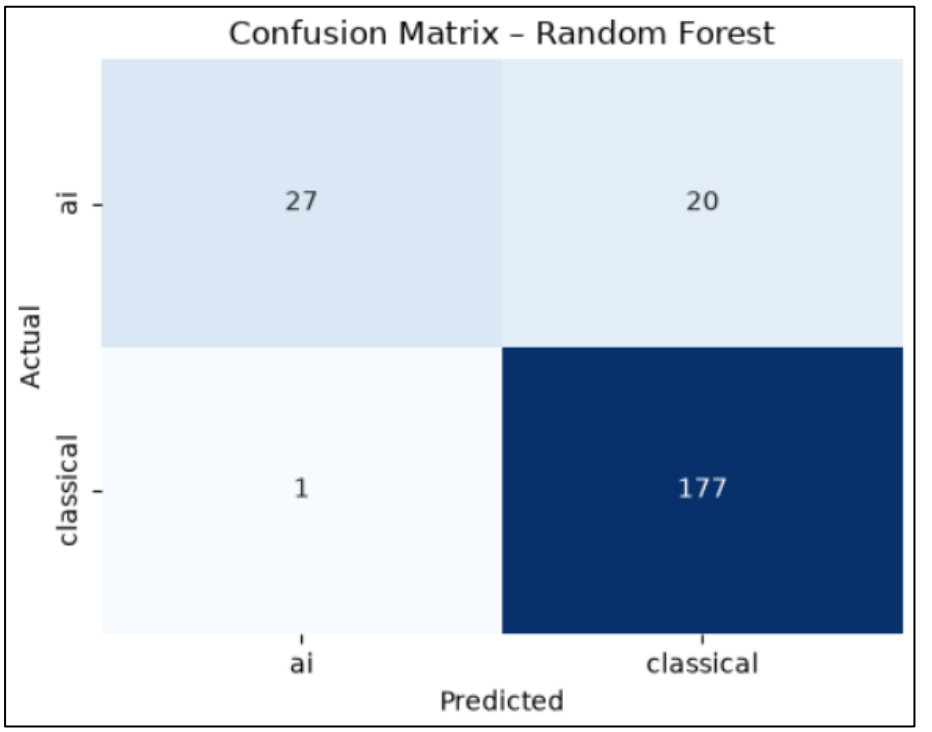


Figure 10: Confusion matrix – Random Forest

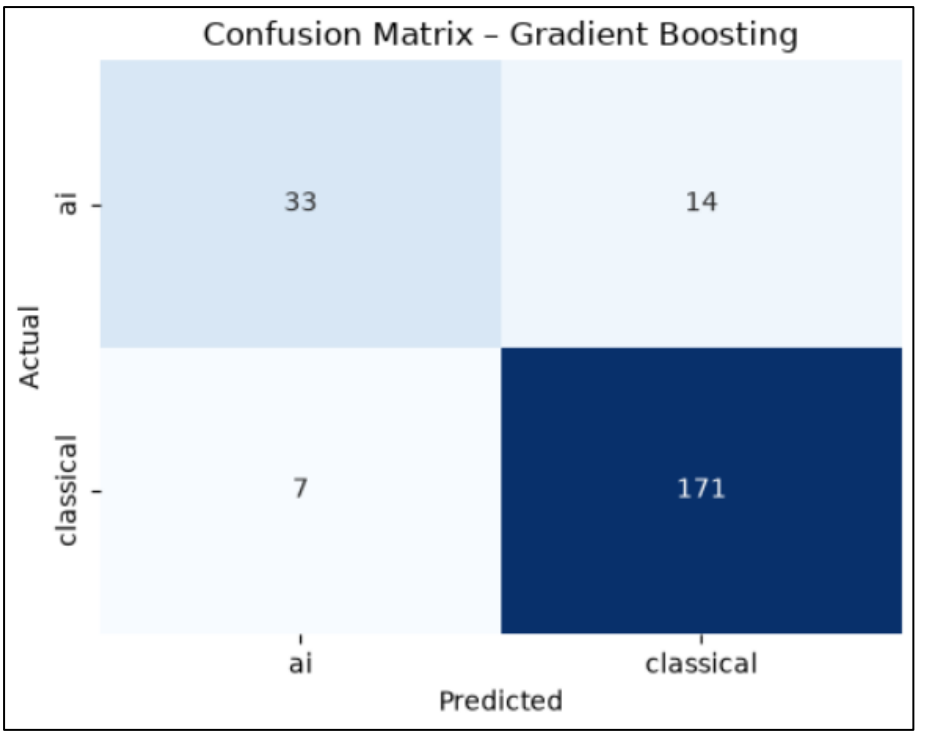


Figure 11: Confusion matrix – Gradient Boosting.

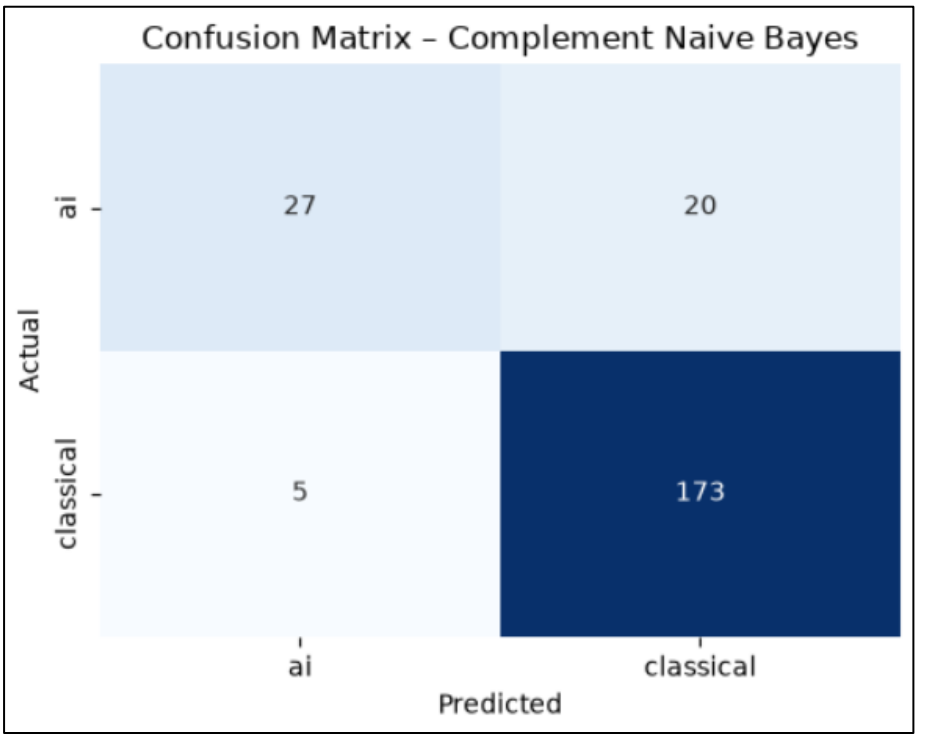


Figure 12: Confusion matrix – Complement Naive Bayes.

2.2) Please see attached Jupyter script.

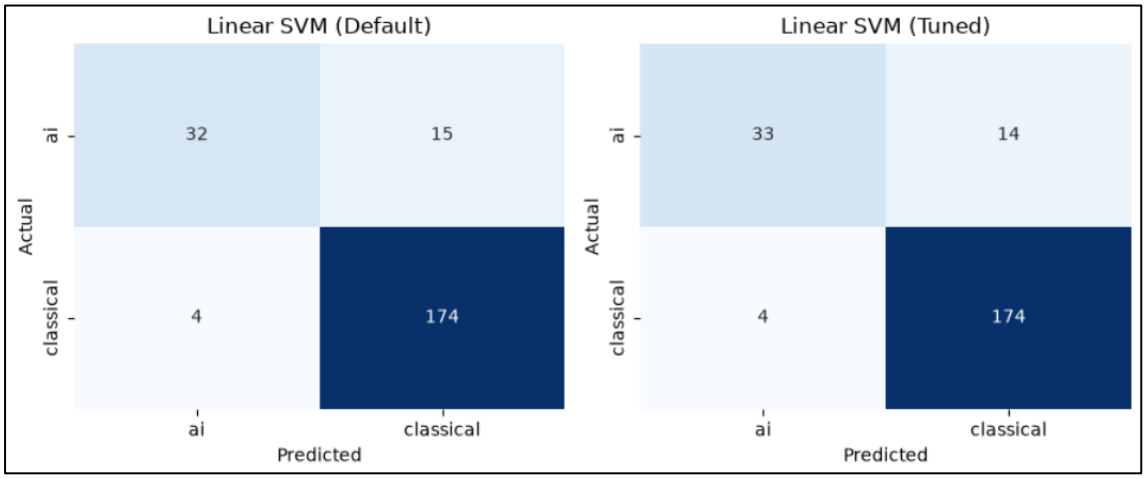


Figure 13: Confusion matrices of the Linear SVM before and after tuning.

The Linear SVM was then tuned with GridSearchCV in a TF-IDF pipeline. The best model (C = 10, bigrams, no feature limit) (Apxml, 2026) reached a CV macro-F1 of 0.849 and a test macro-F1 of 0.87 (Figure 13), and was saved to be used by the app (scikit-learn developers, 2026).

2.3) Please see attached Jupyter script.

The tuned model was deployed as a Streamlit app (Q2.3_app.py) that classifies a pasted abstract as AI or classical research (Figure 14).



Figure 14: Streamlit web app classifying a research abstract.

Question 3

3.1) Please see attached Jupyter script.

The five articles were unzipped and loaded into a DataFrame, and each row contained one article (Figure 15).

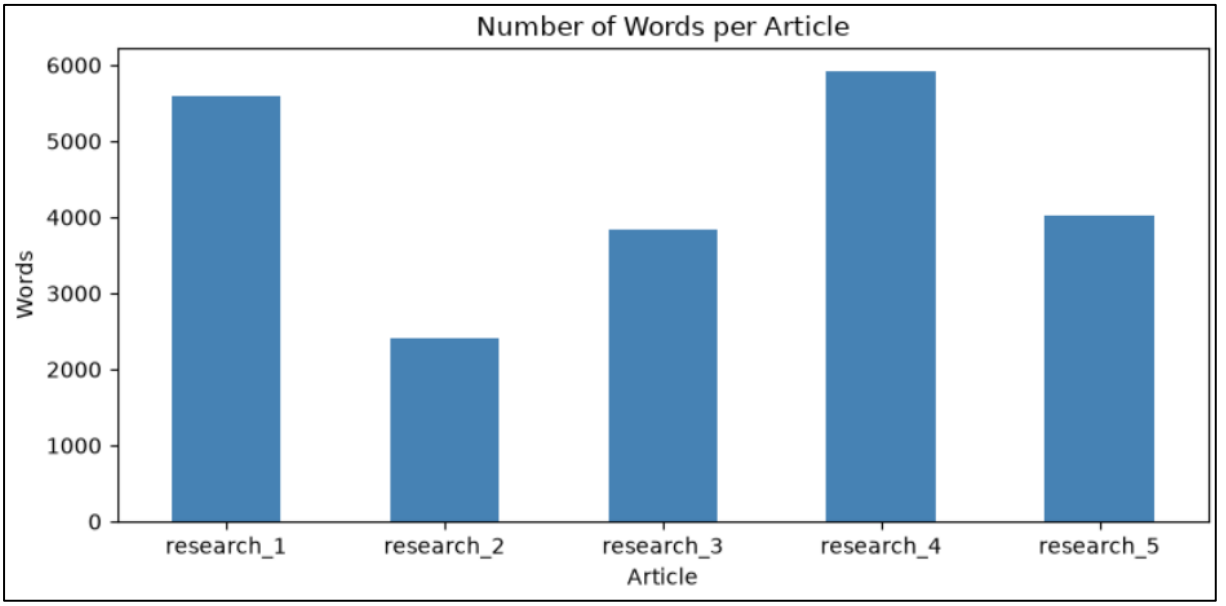


Figure 15: Number of words per article.

3.2) Please see attached Jupyter script.

Duplicate text, page headers, citations, figure labels, and non-English characters were removed. The text was also tokenised and stopwords were removed and the words were lemmatised (Figure 16).

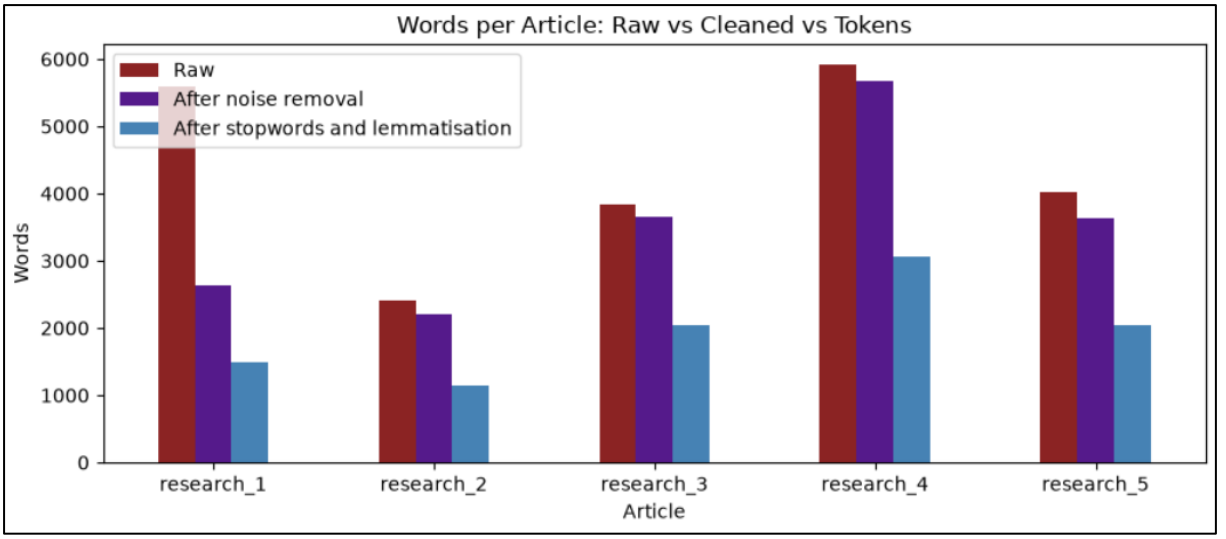


Figure 16: Words per article: raw, after noise removal, and after stopword removal and lemmatisation.

3.3) Please see attached Jupyter script.

Article 1 was summarised with frequency-based extractive summarisation, and it kept the five highest scoring sentences.

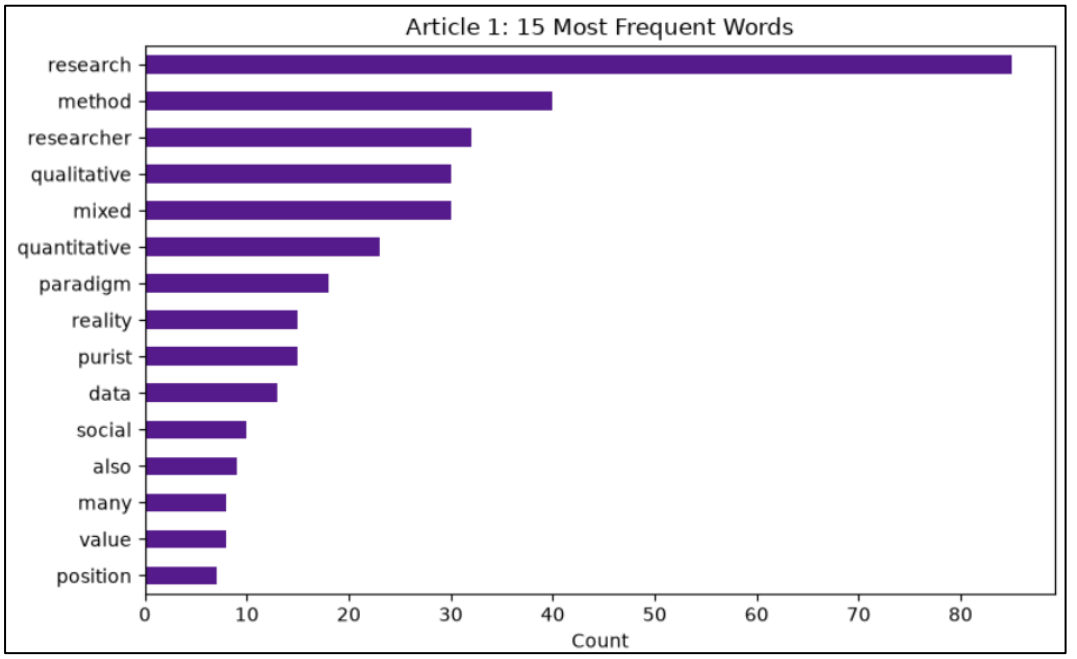


Figure 17: Article 1: 15 most frequent words

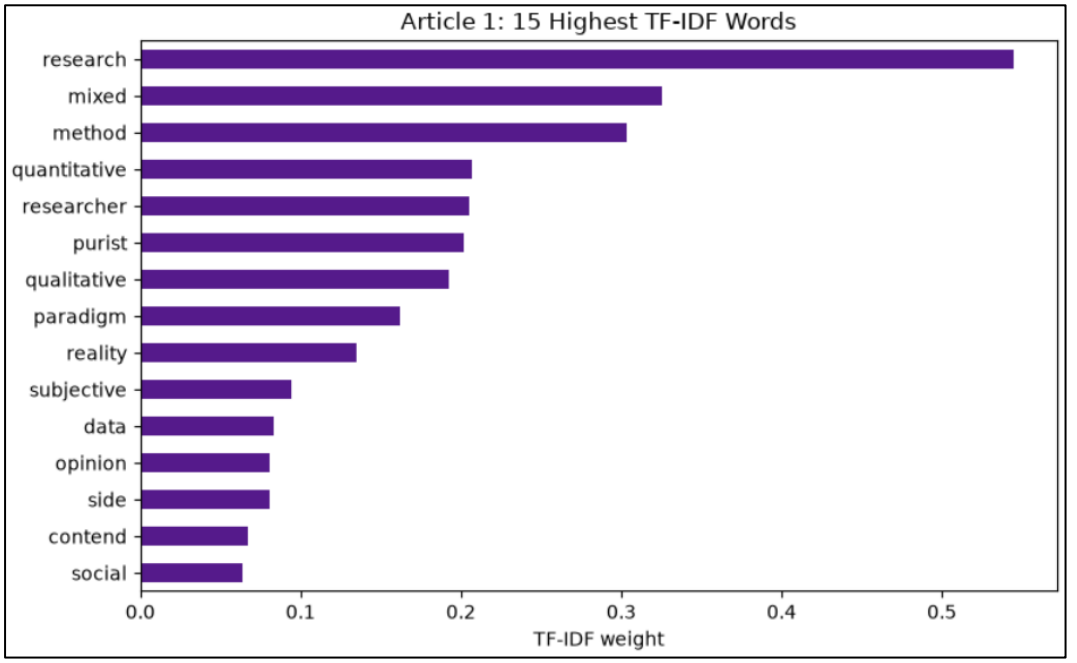


Figure 18: Article 1: 15 highest TF-IDF words.

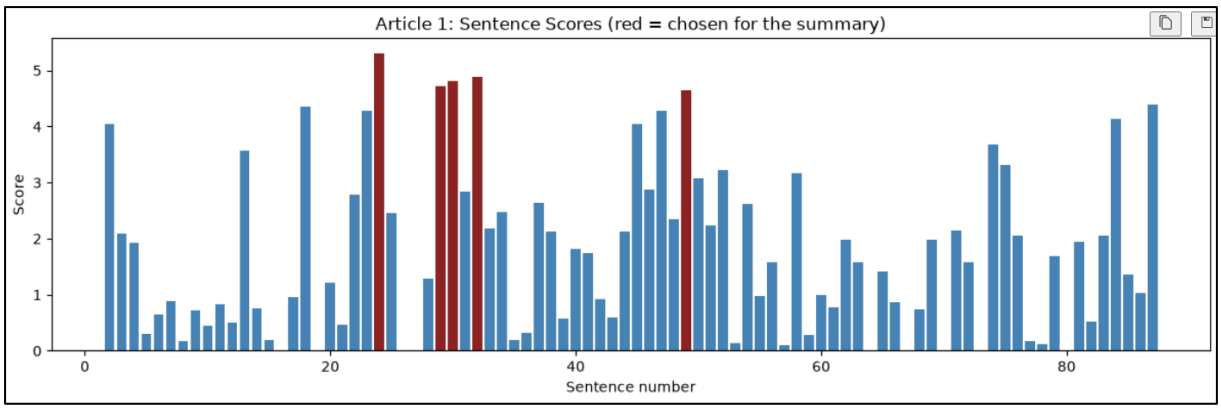


Figure 19: Article 1: sentence scores (red = chosen for the summary).



Figure 20: Article 1: word cloud

3.4) Please see attached Jupyter script.

Articles 2 to 5 were summarised together with TextRank and it kept the top two sentences per article.



Figure 21: Articles 2–5: word cloud.

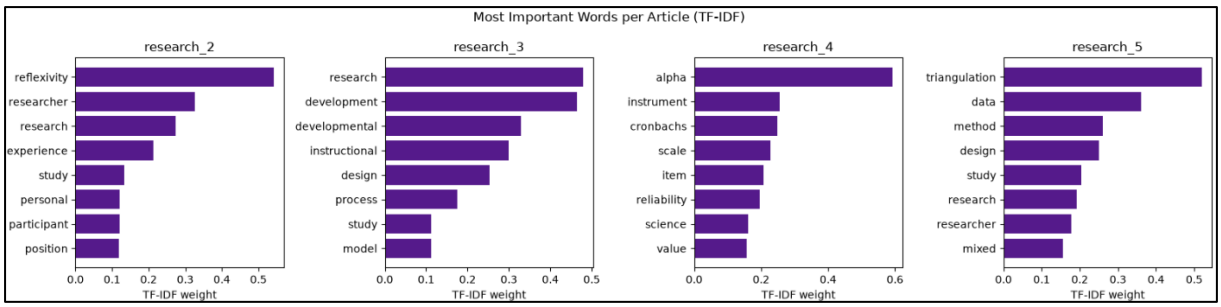


Figure 22: Most important words per article (TF-IDF), articles 2–5.

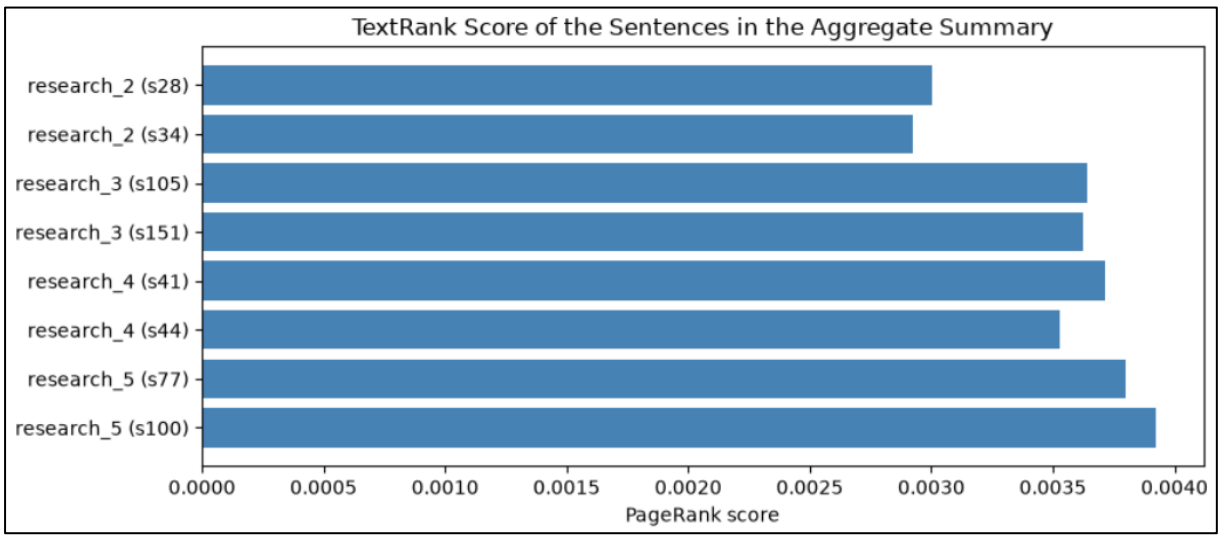


Figure 23: TextRank scores of the sentences in the aggregate summary.

Question 4

4.1) Please see attached Jupyter script.

The articles were split into 100-word passages and LDA was trained for k = 2 to 10. It was found that k = 4 gave the highest coherence score of 0.7517 (Figure 24), as a result, four topics were used (Zvornicanin, 2025).

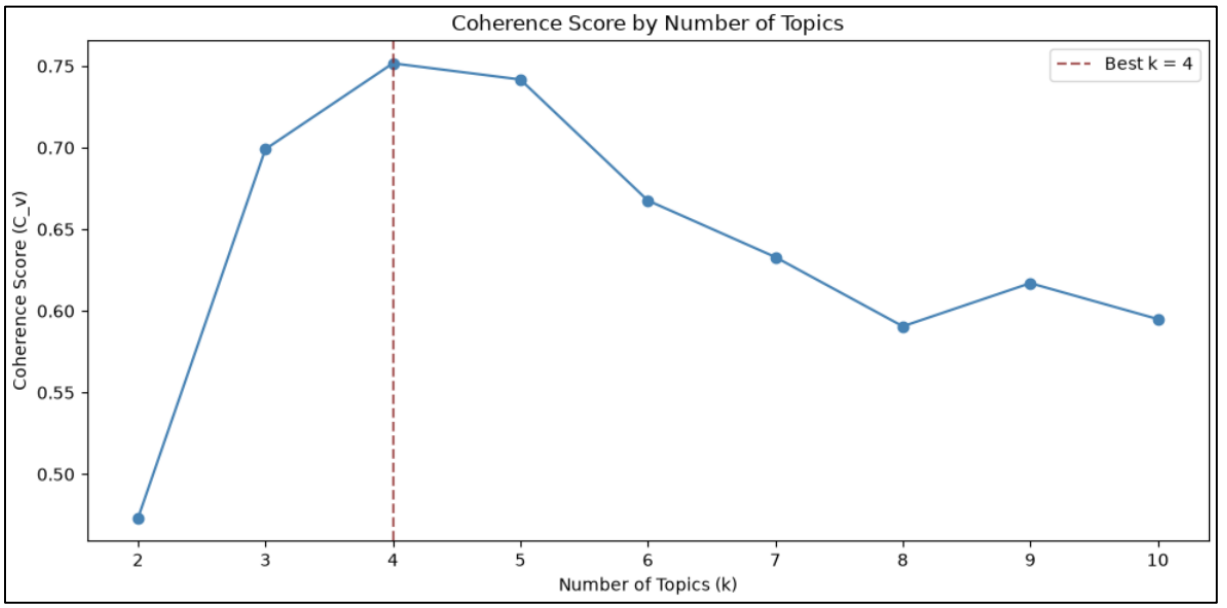


Figure 24: C_v coherence score by number of topics (k).

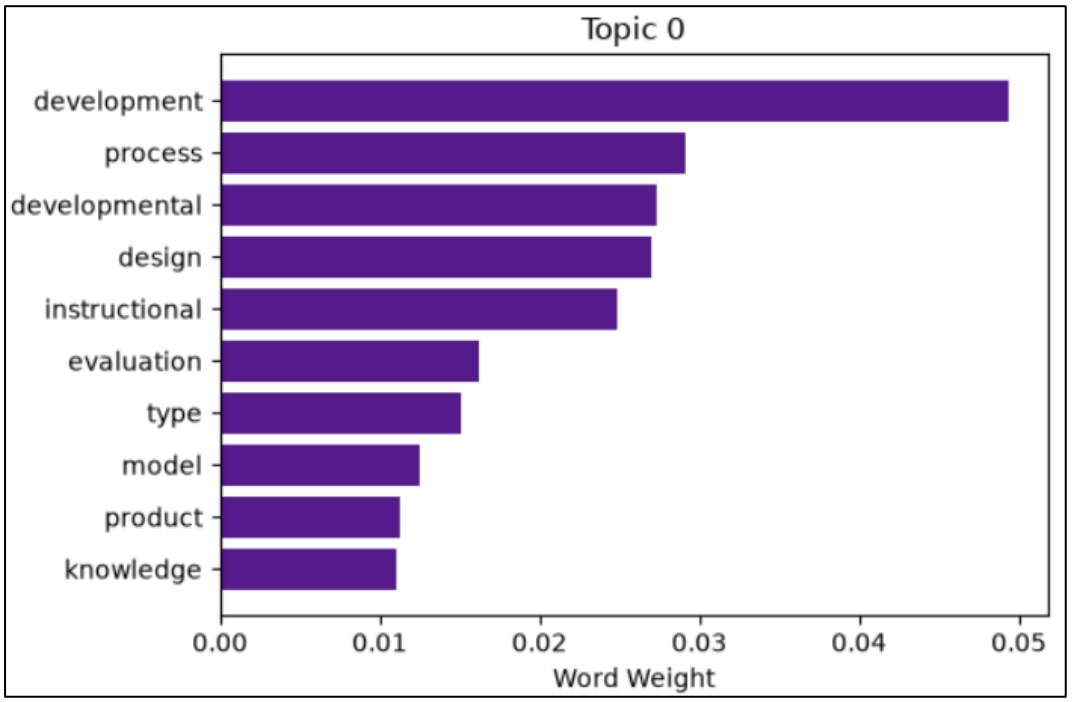


Figure 25: Topic 0 top words.

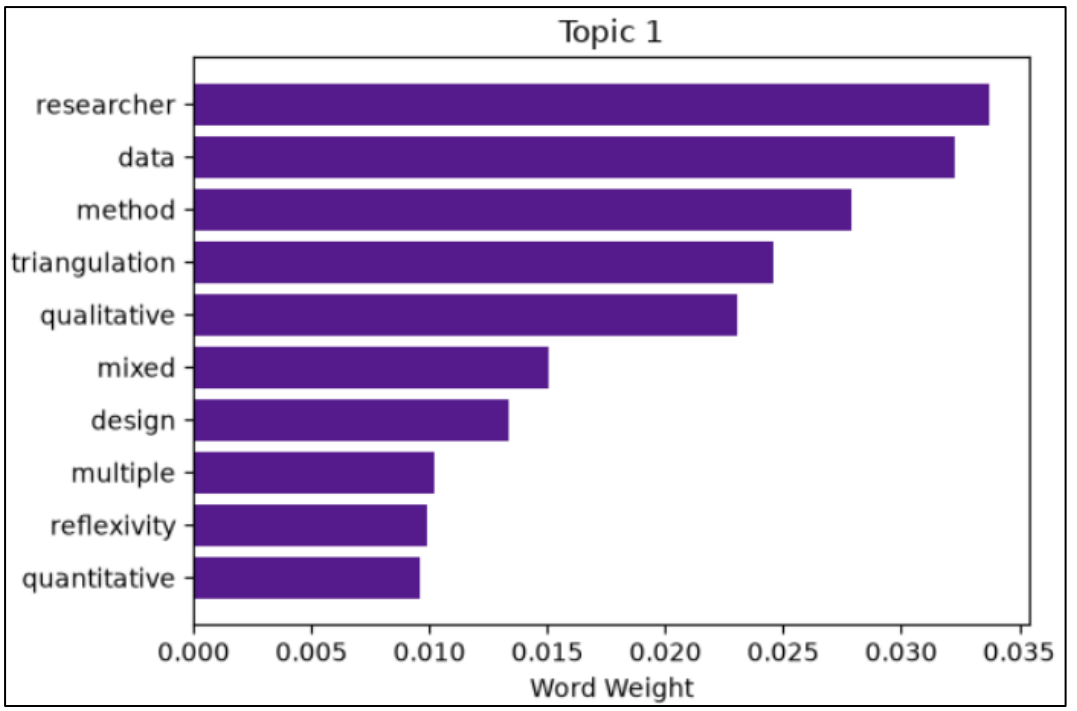


Figure 26: Topic 1 top words.

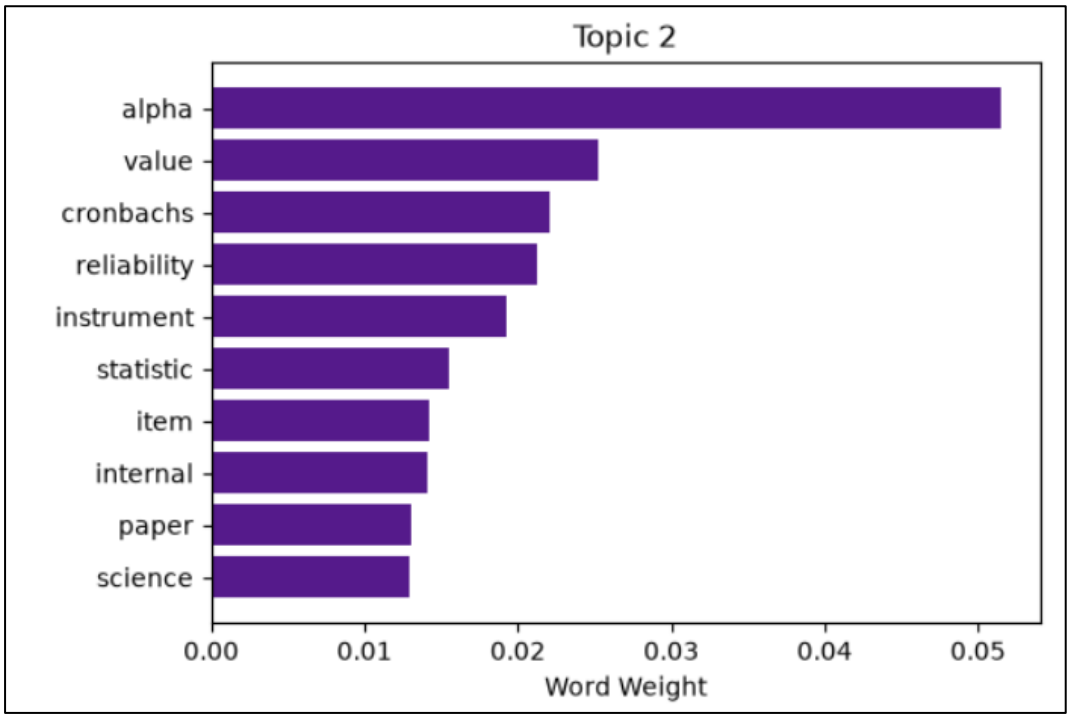


Figure 27: Topic 2 top words.

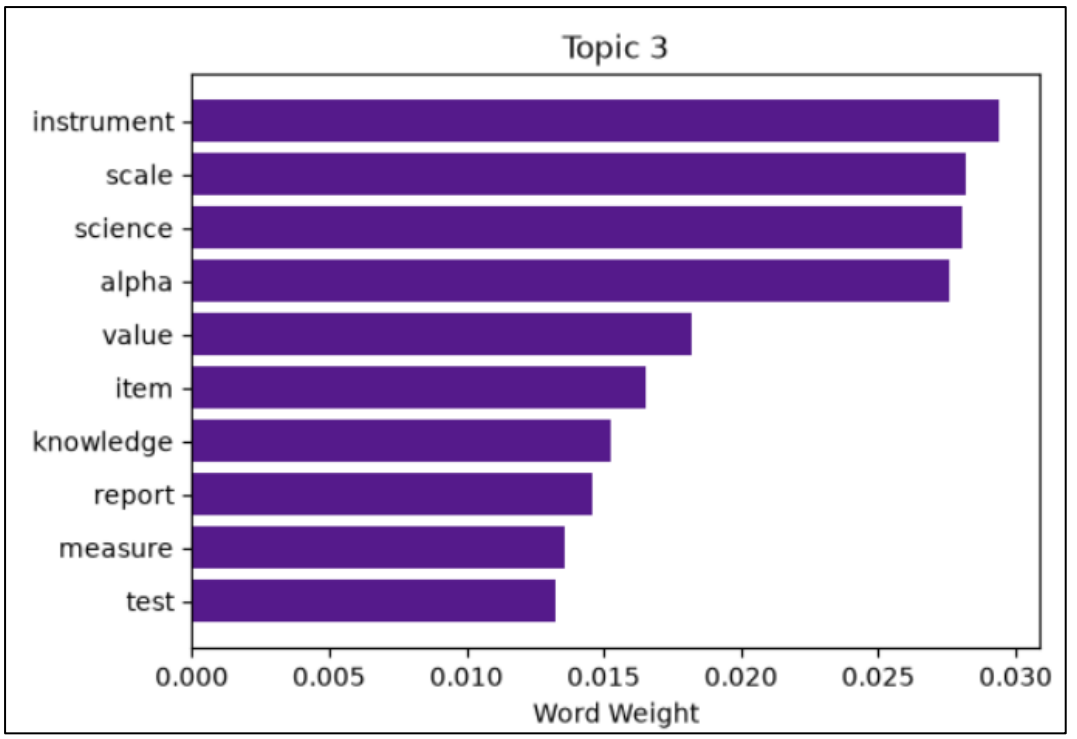


Figure 28: Topic 3 top words.

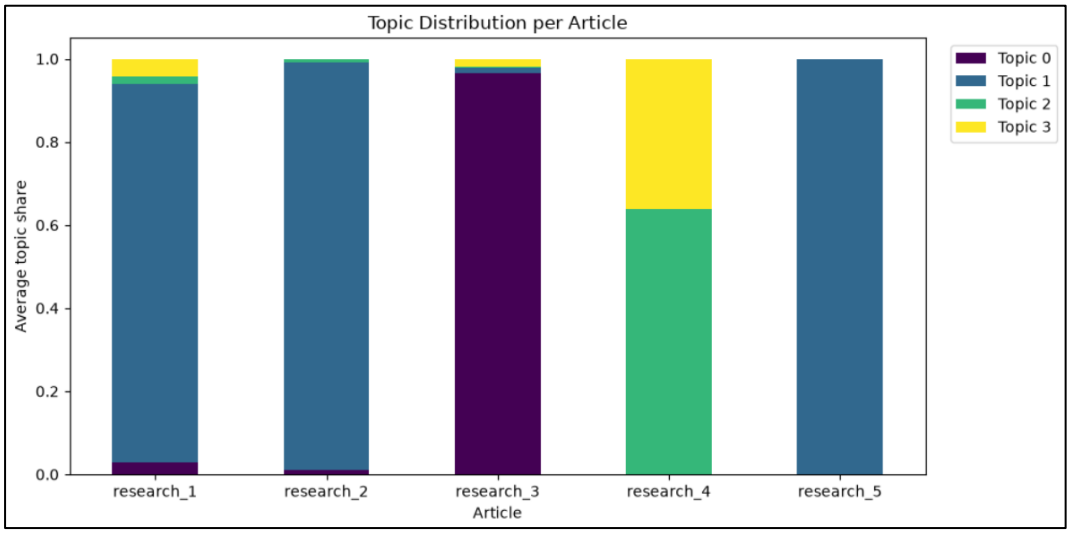


Figure 29: Topic distribution per article.

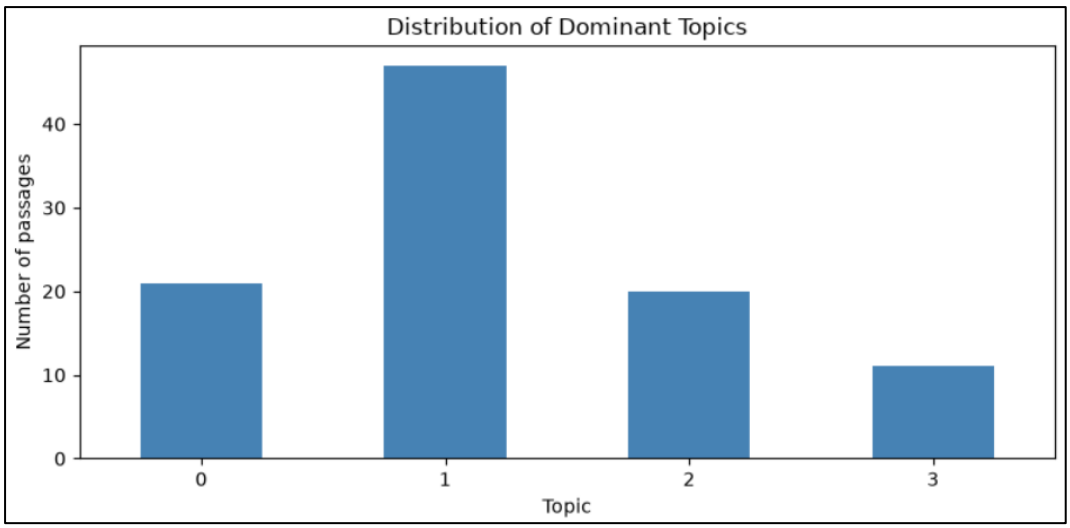


Figure 30: Number of passages per dominant topic.

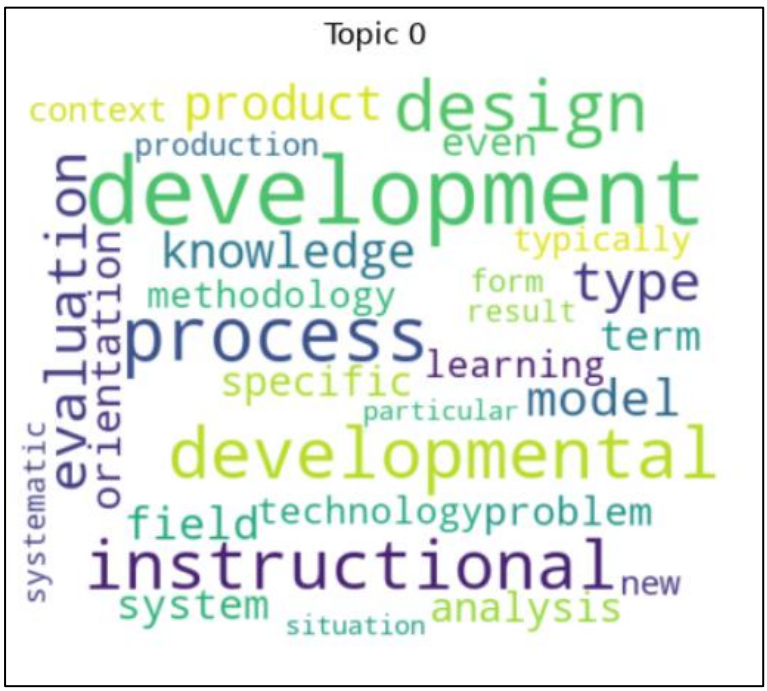


Figure 31: Topic 0 word cloud.



Figure 32: Topic 1 word cloud.

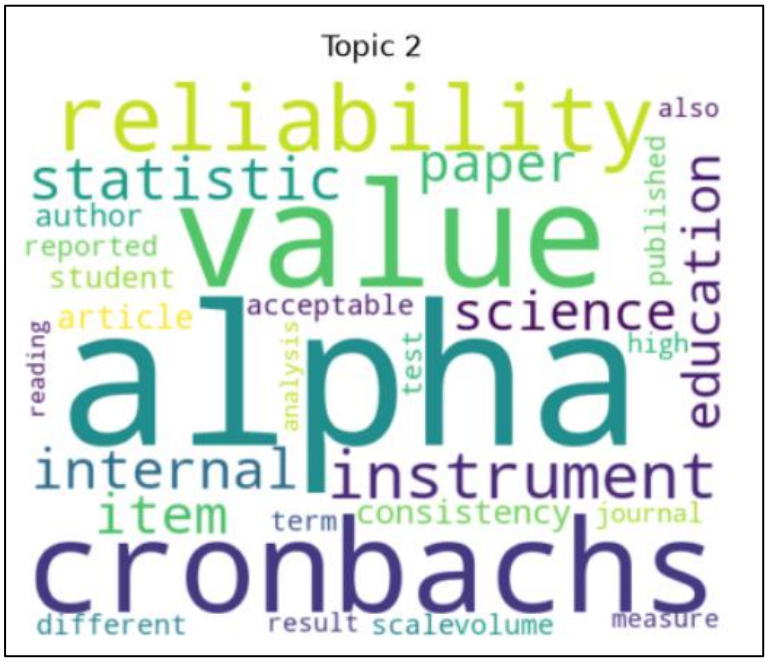


Figure 33: Topic 2 word cloud.

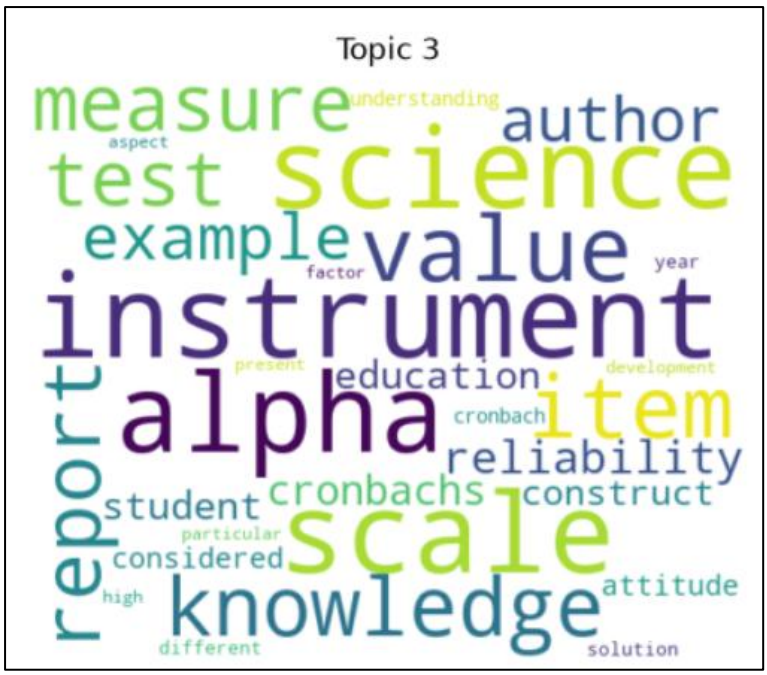


Figure 34: Topic 3 word cloud.

4.2) Please see attached Jupyter script.

The topics were visualised with pyLDAvis, where each bubble is a topic and the bigger the bubble, the more prevalent the topic is (Figure 35) (Prabhakaran, 2026).

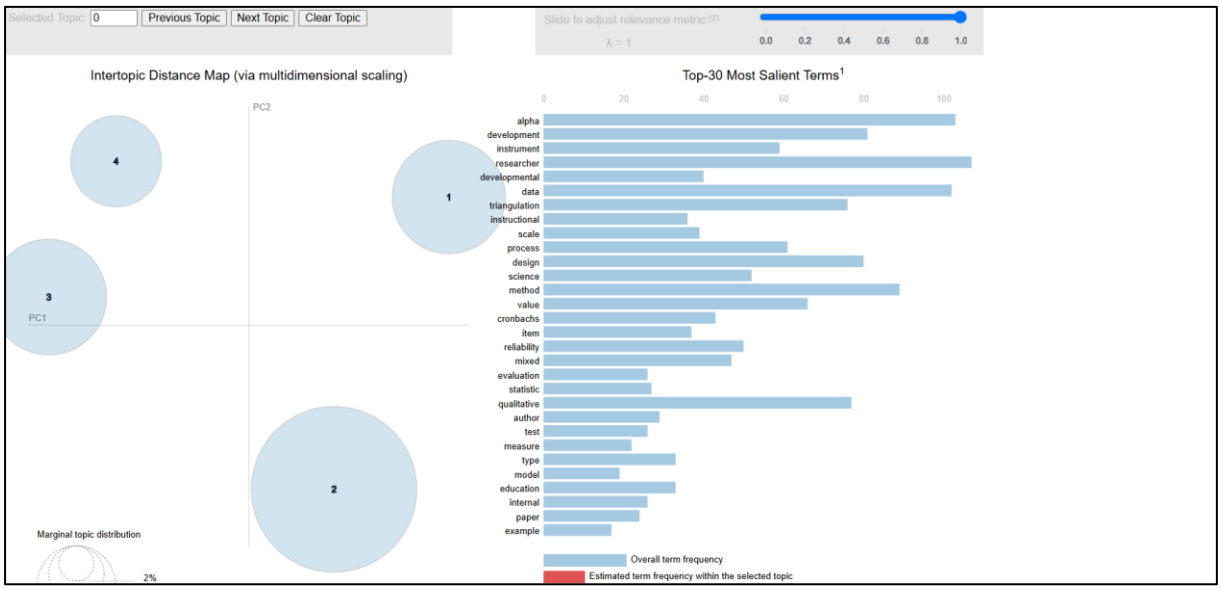


Figure 35: pyLDAvis intertopic distance map and most salient terms.

4.3) Please see attached GitHub Repository link:

https://github.com/Jess871/ITNPA4-Project

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